

INVESTIGATING MECHANISMS OF NOCTURNAL OZONE IN NORTHWEST INDIANA AND CHICAGOLAND

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INTRODUCTION

- Surface ozone is a secondary pollutant produced through photochemical reactions involving NO_x and VOCs.
- Elevated ozone concentrations pose risks to human health, vegetation, and ecosystems.
- Ozone typically decreases after sunset, making persistent nighttime concentrations unusual.

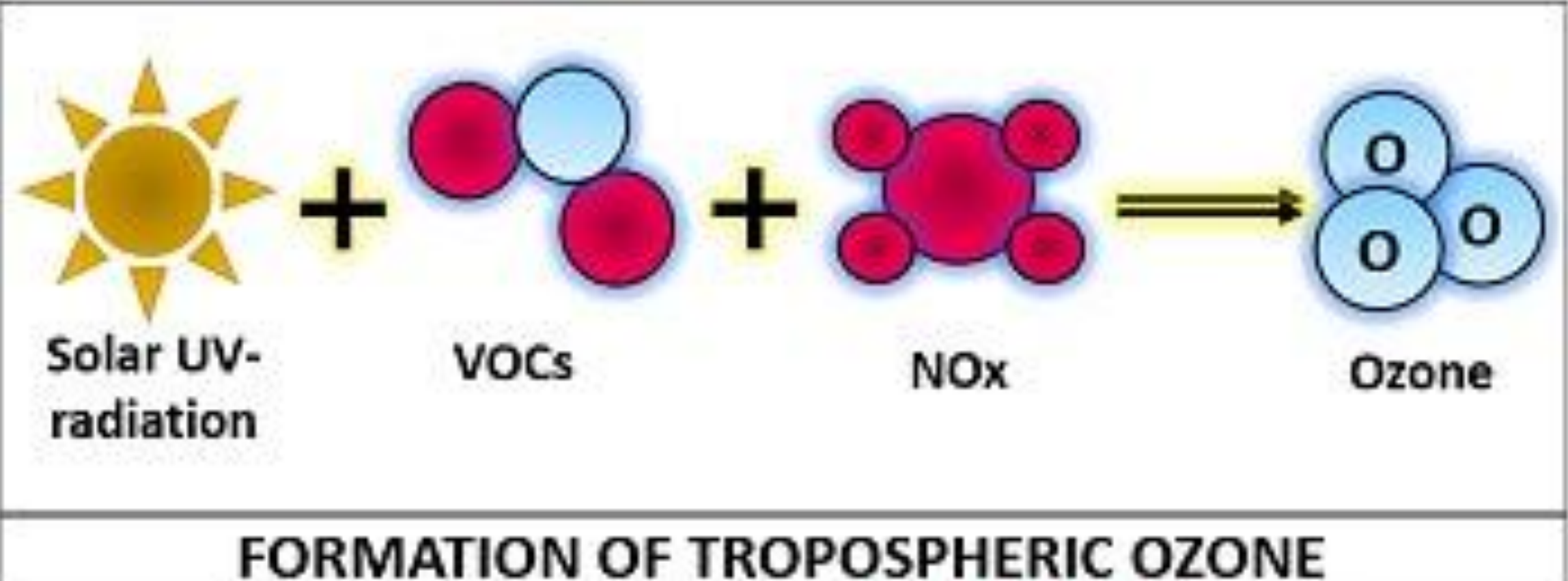


Figure 1: Formation of Tropospheric (surface) ozone. <https://biologyreader.com/ozone-formation.html>

MOTIVATION

- Investigate why elevated ozone concentration persisted overnight in Valparaiso, IN.
- Determine whether transported pollution from the Chicago metropolitan area contributed to nighttime ozone.
- Improve understanding of meteorological conditions responsible for prolonged ozone episodes.

METHODS

- Ozone measurements from the VU Ozone Garden, EPA monitoring stations, and CROCUS sensors were used to characterize the evolution of ozone concentrations and compare observations across monitoring networks.
- Weather conditions derived from ASOS observations and synoptic analyses were examined to identify the meteorological conditions associated with the ozone event.
- HYSPLIT modeling and ERA5 meteorological analyses were used to identify transport pathways and evaluate the influence of regional meteorological conditions on ozone transport.
- TEMPO satellite observations of tropospheric NO₂ and ozone were analyzed to characterize the spatial distribution of air pollutants and assess regional transport and the influence of the Lake Michigan lake-breeze circulation.

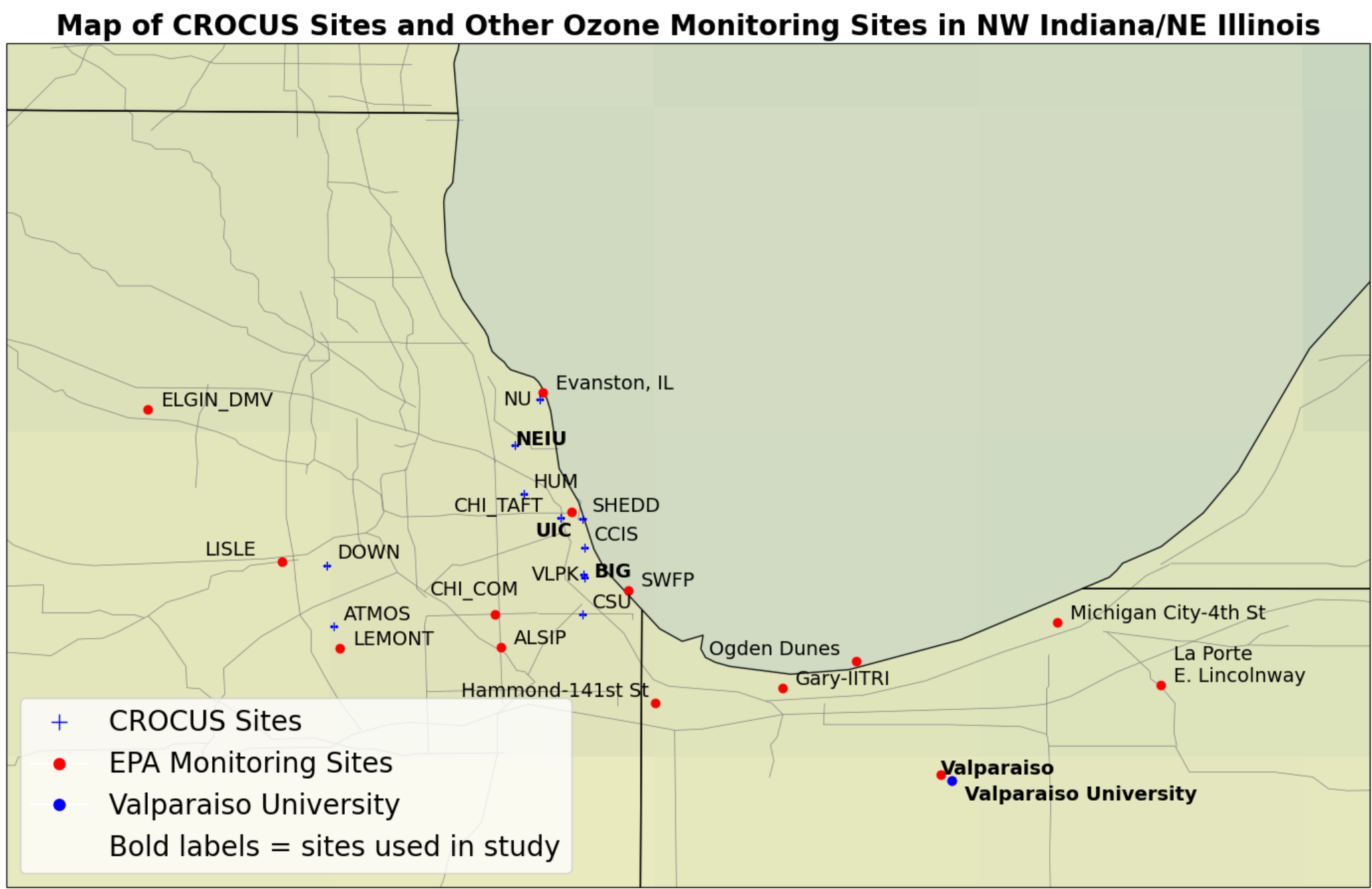


Figure 5: Map of available Ozone monitoring sites in the study region, bolded are sites of interest.

CONCLUSIONS

- Above-average temperatures, southwesterly flow, and lake-breeze circulation, favored enhanced ozone production during the study period.
- Valparaiso exhibited persistent nighttime ozone and secondary nocturnal ozone peaks that were not as pronounced at nearby monitoring sites, suggesting that downward mixing from the residual layer played an important role in sustaining elevated nighttime ozone concentrations after sunset.
- Regional transport, supported by meteorological analyses and HYSPLIT backward trajectories, together with nocturnal boundary-layer processes, likely contributed to the elevated nighttime ozone concentrations observed at Valparaiso.

FUTURE RESEARCH

- Further investigate the role of Lake Michigan and lake-breeze circulations in regional ozone transport.
- Deploy ozonesondes to characterize vertical ozone profiles.
- Examine why periods of persistent southerly flow are associated with elevated ozone concentrations.
- Expand the analysis to additional Midwestern monitoring sites to identify similar nocturnal ozone events.
- Further ozone garden investigation of space is planned for the end of summer 2026.



Figure 2: Valparaiso University (VU) Ozone Garden, Credit: Dr. Katelyn Barber.

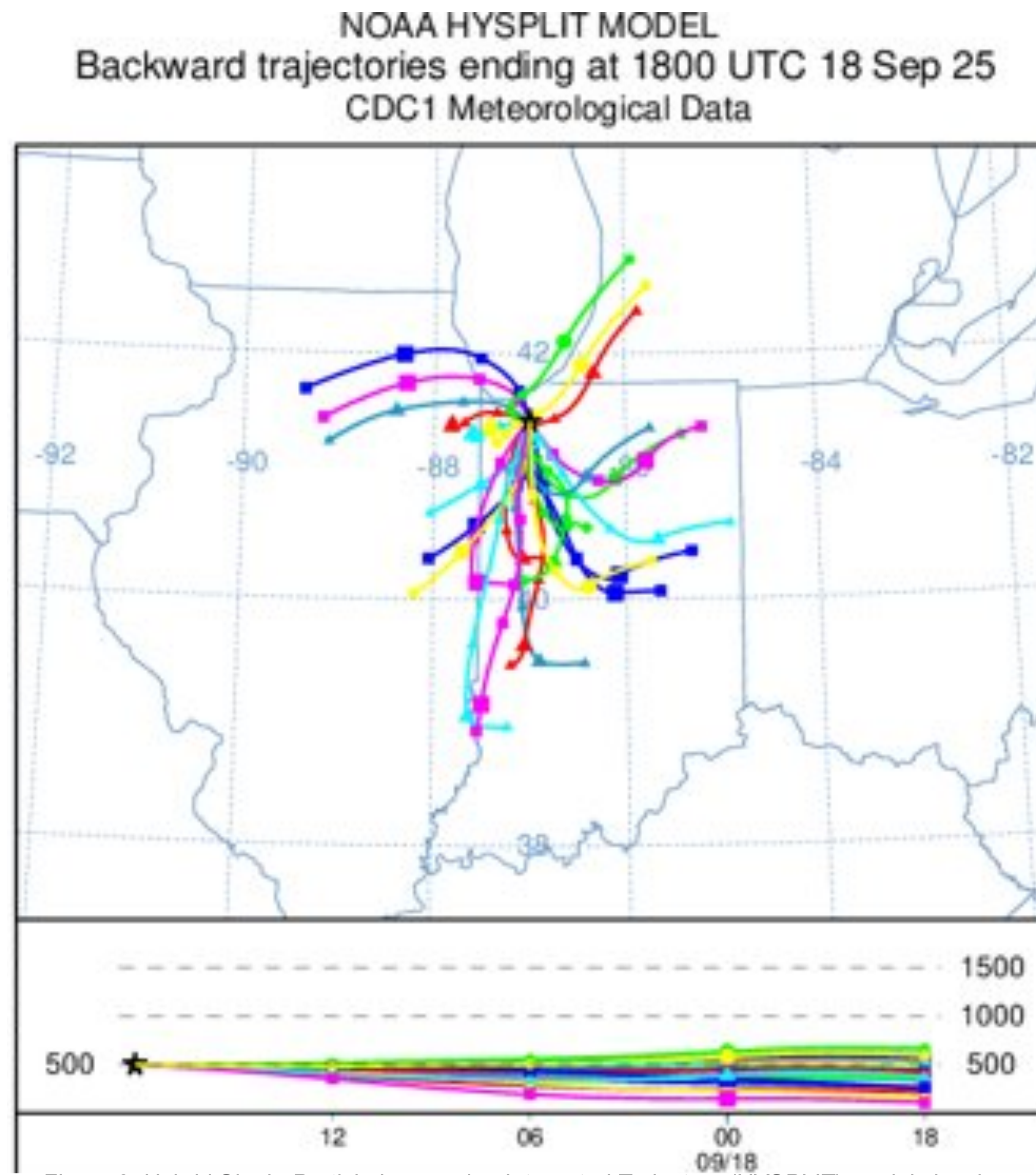


Figure 3: Hybrid Single-Particle Lagrangian Integrated Trajectory (HYSPLIT) model showing where air masses originated from over Valparaiso, IN at 18 Z (1 pm LT).

SYNOPTICS-STUDY PERIOD

- Above average temperatures for September (up to 31.7 °C) and mostly cloud-free conditions promoted strong daytime photochemical ozone production across northern Illinois.
- The prevailing synoptic flow governed regional transport, while an afternoon Lake Michigan lake breeze shifted nearshore winds onshore and advected cooler lake air inland along the Chicago shoreline.
- Tropospheric TEMPO NO₂ observations showed elevated precursor concentrations over the Chicago metropolitan area, while enhanced total column ozone extended across northern Illinois and northwest Indiana.
- HYSPLIT backward trajectories indicated that air masses reaching Valparaiso originated from multiple directions, reflecting complex regional transport associated with the prevailing synoptic pressure pattern.
- Weak nighttime winds and evening soundings were consistent with the development of a stable nocturnal boundary layer, which likely reduced vertical mixing after sunset.

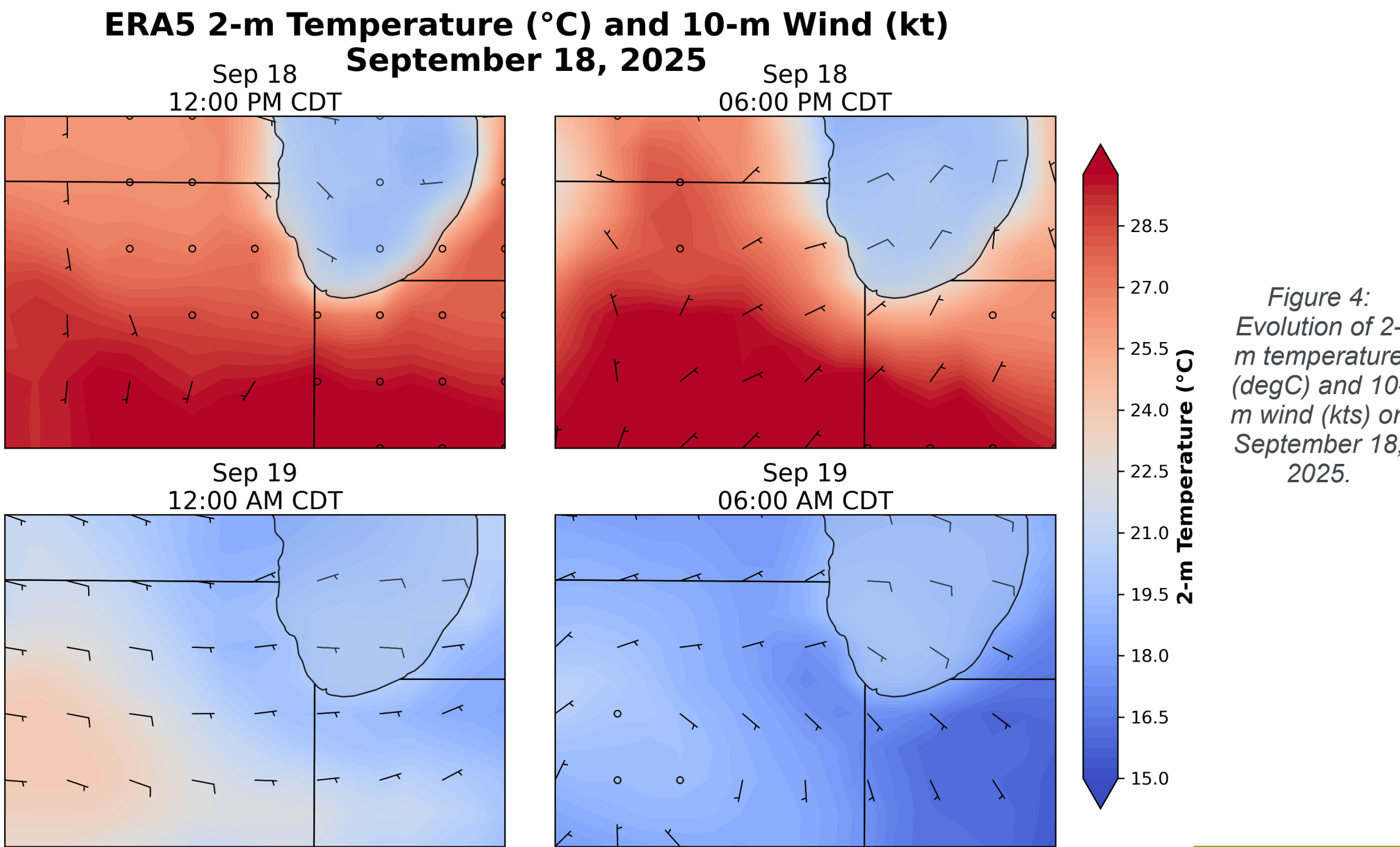


Figure 4: Evolution of 2-m temperature (degC) and 10-m wind (kts) on September 18, 2025.

PRESENCE OF NOCTURNAL OZONE

- Ozone concentrations at the VU Ozone Garden remained elevated after sunset, rather than exhibiting the typical nocturnal decline observed at most surface monitoring sites.
- Nighttime ozone concentrations at VU exceeded those measured at several nearby CROCUS monitoring sites, indicating localized differences in nocturnal ozone behavior.
- The persistence of elevated nighttime ozone is consistent with regional transport and subsequent downward mixing of ozone from the residual layer, suggesting that transport and boundary-layer processes contributed to the nocturnal ozone event.
- The regional wind field supported transport of ozone-rich air into northwest Indiana during the day, allowing elevated ozone concentrations to persist into the evening.

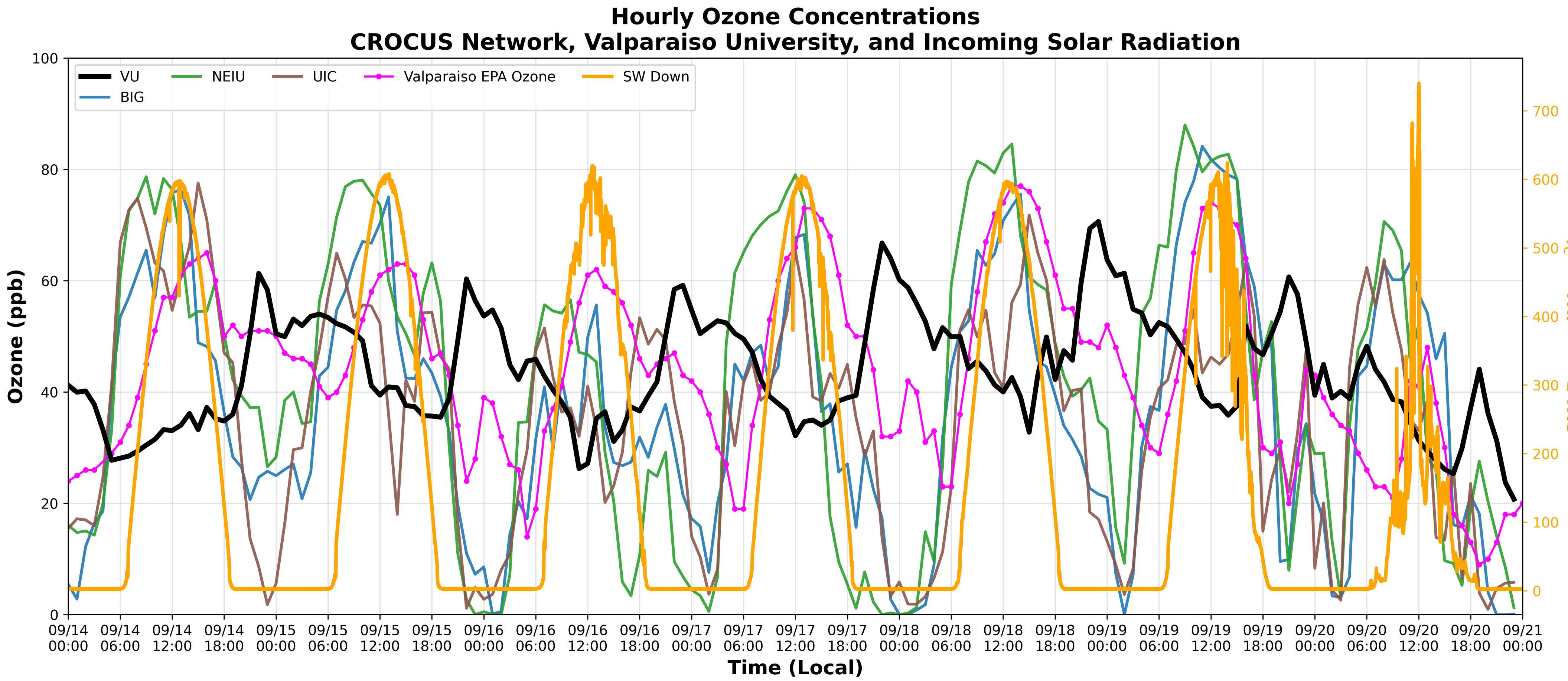


Figure 6: Time series showing when ozone peaks occurred at each site with Shortwave Radiation overlaid.

REFERENCES

- reference dr. barber for her awesome photography skills
- CROCUS Data
- biology reader
- EPA
- Valpo Data
- HYSPLIT
- Wyoming sounding archive

ABSTRACT

In 2025, Valparaiso University (VU) joined the National Ozone Garden Network. This nationwide program uses ozone-sensitive plants to monitor the effects of ozone pollution alongside an in situ ozone monitor. During the summer of 2025, ozone concentrations exceeded the U.S. Environmental Protection Agency (EPA) National Ambient Air Quality Standard of 70 ppb only nine times, with the highest concentration occurring on 18 September, 2025. This peak occurred during a regional ozone episode from 14-21 September, 2025, when multiple Air Quality Action Days were issued across Northwest Indiana and the Chicago metropolitan area. To investigate the meteorological conditions and transport processes responsible for elevated ground-level ozone concentrations during this event, surface ozone observations from the VU Ozone Garden, EPA monitoring stations, and the Community Research on Climate and Urban Science (CROCUS) sensor network were combined with meteorological observations from Automated Surface Observation System (ASOS) stations, and synoptic weather analyses. By comparing ozone concentrations, wind patterns, and regional weather conditions, the influence of ozone transport from the Chicago metropolitan area and the role of lake breeze circulations in producing elevated ground-level ozone concentrations in Valparaiso can be determined. Understanding the factors that contributed to this high-ozone event will improve knowledge of ozone episodes in Northwest Indiana and support future air quality forecasting and public health awareness.

